

KV Cache Optimization in LLMs: A Taxonomy and Decision Framework for Efficient Inference

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Abstract—Large Language Models rely on the Key-Value (KV) cache to save on computing projections of previously generated tokens in autoregressive inference. Nevertheless, the KV cache scales linearly with the context size and becomes a major limiting factor for GPU memory, memory bandwidth, and inference latency. In this work, we propose a taxonomy based on bottleneck identification and analyze ten KV cache optimizations techniques belonging to four different classes attention level redesign, quantization, compression, and memory management in a unified way with respect to a set of common metrics. The analysis is based on a reproducible benchmarking framework, kvbench, measuring the peak GPU memory, CPU resident-set size, wall-clock latency, throughput, and analytical KV cache size on an identical set of prompts, seeds, and (context, batch) settings. Three optimizations whose kernels are proprietary (NVFP4, MiniKV, xKV) are simulated by high-fidelity analytical models. For the five techniques that run a real model directly (Baseline FP16, INT8, FP8, GQA, and low-rank compression), the accuracy trade-off is verified with a measured WikiText-2 perplexity evaluation (Section V-D), while the three simulated techniques retain the memory-saving figures reported by their original publications. The proposed framework and the comparison serve as a decision framework for choosing KV cache optimization techniques in various scenarios of memory, latency, and accuracy constraints.

Index Terms—KV cache, large language models, inference optimization, quantization, compression, memory management, attention, benchmarking.

I. INTRODUCTION

Recently, Large Language Models (LLMs) have become a core foundation in AI-driven applications due to the sole reason of powerful understanding and generation of natural language. Unlike traditional AI systems, LLMs provide great flexibility and handle wide range of tasks with enhanced adaptability [1]. To manage active information, LLMs utilize a short-term memory technique known as the context window. Within this window, the model processes previous conversational turns, external documents, and its own prior outputs. To generate logical text, LLMs rely on an autoregressive process which generates tokens one by one based on all previous tokens to understand the broader context of the sequence [2].

A main challenge for LLM deployment is how models manage, access and retain information across interactions. Due to this autoregressive behaviour, the computing cost of a model increases as it recalculates the mathematical relation of previous tokens each and every time a new token is generated [2]. This leads to computational overhead within the context window. In some cases input context can be tens and hundreds of times longer than the output it generates [3].

To overcome the limitation of recomputation of past token, the key-value (KV) cache is used which stores the intermediate attention states (Keys and Values) of past tokens in memory during the processing phase [4]. So when new tokens are generated it does not recalculates but reuse the prior computation. Resulting in reduction of latency and make autoregressive generation more efficient.

Although the KV cache has improved the inference efficiency of Large Language Models (LLMs), its memory footprint grows linearly with context length which increased memory consumption, introduce bottlenecks in GPU memory capacity, memory bandwidth, inference throughput, and latency, especially for long-context and multi-user inference [4].

As LLMs can handle longer context, different methods has been introduced to improve and optimize performance of KV cache and most of them focuses on single technique. Limited work has been published which organize these methods, making it difficult to understand their concepts, how they work, their advantages and their limitation. This report organize the KV cache and the different optimization methods used in modern LLMs and covers techniques such as KV cache compression, quantization, token selection, offloading, and memory management. These techniques are then compared based on memory usage, computational cost, inference speed and their applications and provide appropriate guideline for selecting suitable methods under different memory, latency, and computational constraints.

II. LITERATURE REVIEW

A. Background

KV cache is crucial for LLM deployment as it reduces latency by avoiding recomputing past tokens, lowering operational costs with less compute requirements, and enables deployment on resource-constrained devices by managing large memory and optimizing data transfers. KV Cache stores intermediate key which represents identity of token and value vectors which represents actual content of that token from previous tokens to avoid redundant computation during generation [4]. And when new token is generated, the key and value pair are appended to the cache. The KV cache size is expressed as:

$$KV_{\text{per token}} = 2 \times H \times L \times B \times D \quad (1)$$

In Equation (1), [4] H is the number of attention heads; D the dimension of each head; L the number of transformer layers; B the bytes per element. These parameters are fixed for a given model, the KV per token is constant for a model.

$$KV_{\text{cache size}} = KV_{\text{per token}} \times \text{Context Length} \quad (2)$$

Equation (2) shows that the total KV-cache memory increases linearly with the context length. As modern LLMs support context windows containing tens or hundreds of thousands of tokens, the KV cache becomes a major bottleneck in terms of GPU memory consumption, memory bandwidth, and inference latency [2], [4].

Each optimization techniques reduces either one of the terms of Equation (1) to optimize and improve efficiency of KV cache. To deal with this bottleneck, there are different techniques which can be categorized into attention-level redesign, quantization, compression, and memory management.

B. Attention Level Techniques

Ainslie et al. [5] introduces Grouped-Query Attention (GQA), which groups multiple query heads to share same key and value, as shown in Fig. 1 reducing cache size while losing little output quality. GQA stays in between two attention techniques: standard Multi-Head Attention (MHA), where each query head has its own key and value, and Multi-Query Attention (MQA), where all query heads share a single key and value.

The result from the experiment shows that GQA inference speed is close to that of Multi-Query Attention (MQA) and maintains performance similar to that of standard Multi-Head Attention. The authors also introduce an effective way to upgrade existing MHA models into GQA models using mean pooling, requiring only a small amount of time and cost to adapt [5].

C. KV Cache Quantization

KV Cache Quantization is the optimization technique which is used to reduce memory requirement to store key and value tensors. Normally, these tensors are stored in high precision format usually FP16, instead of that these cached tensors

are represented in lower precision format such as FP8, FP4. This helps in decreasing memory footprint and allows longer context length while maintaining model accuracy [2].

To reduce memory usage, multiple studies have introduced different KV cache quantization methods. FP8 KV cache is widely used because of its balance in memory savings and in model accuracy. A newer method, NVFP4 KV cache has been introduced by NVIDIA, which stores key and value tensors in 4-bit floating point format, reducing KV cache memory by half compared to that of FP8, allowing longer context lengths, supports more users, and lowers memory bandwidth during decoding. Cached values are dequantized for attention computation and quantized again before being stored [2].

Another method is MixQuant [3] which uses mixed precision quantization by providing different precision levels to different cached tensors based on their importance. This reduce memory while maintaining better accuracy than using low level precision for all tensors.

MiniKV [6] improves efficiency of KV cache by combining 2-bit quantization and removes less important cached tokens. It not only reduce size of KV cache through quantization, it also frees memory by keeping only the most useful information. MiniKV reduces KV cache memory by more than 80% while maintaining accuracy and providing faster inference than methods that use only quantization.

TABLE I
COMPARISON OF KV CACHE QUANTIZATION TECHNIQUES.

Method	Precision	Main Benefit	Trade-off
FP16	16-bit	Highest accuracy	Large memory usage
FP8	8-bit	Lower memory	Slight accuracy loss
NVFP4	4-bit	Smaller cache	Quantization overhead
MixQuant	Mixed precision	Better accuracy-memory trade-off	Higher complexity
MiniKV	2-bit + adaptive eviction	80% compression	Custom kernels

Table I summarizes recent KV cache quantization techniques based on the findings in [2], [3], [6].

Although KV Cache Quantization reduces memory usage, it reduces generation quality for a longer context because of using low precision tensor format. As a result, current research focuses on finding the right balance between memory efficiency and model accuracy to enable the practical deployment of modern LLMs [2], [3], [6].

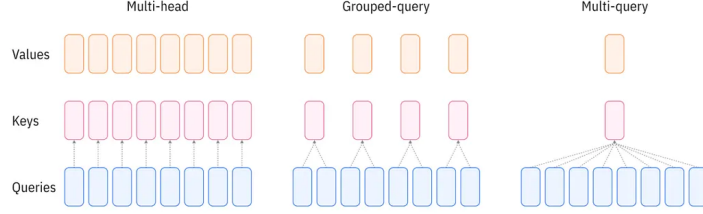


Fig. 1. An structural overview of the Grouped-Query Attention (GQA) mechanism compared alongside Multi-Head Attention (MHA) and Multi-Query Attention (MQA) [5].

D. KV Cache Compression

When context length increases, the KV cache occupies a large portion of GPU memory. While quantization reduces the precision of cached tensors, KV-cache compression decreases the amount of information stored within the key-value representations by removing redundancy. It reduces memory consumption while preserving the necessary information for accurate results [1].

Many studies have provided different ways to reduce the size of KV cache. One way is low-rank compression, which stores the key and value information in a smaller form while keeping most important informations. Another method is KV merging, where similar key value pairs are combined which reduce the number of stored entries. Recent studies use multi-level compression by keeping important tokens in better quality while compressing less important or older tokens to save memory [1].

Chang et al. [7] found that KV caches from nearby transformer layers contain similar information. Instead of storing the KV cache for every layer separately, xKV uses Singular Value Decomposition (SVD) to combine similar information from multiple layers into a smaller shared representation. This helps reduce the amount of memory needed while still maintaining good model accuracy. xKV can reduce the KV cache size by up to $8\times$, and when it is combined with 4-bit quantization, the compression increases to $25.6\times$ with only a small reduction in accuracy.

This shows that xKV can be used together with other methods, such as quantization and intra-layer compression. Rather than replacing these techniques, xKV helps improve memory savings when they are used together.

Although KV cache compression uses less GPU memory and allow the model to handle longer input without using too much extra memory, compare to that of normal KV cache. If the data is compressed too much, some important information may be lost, which can reduce the quality of the output.

E. Memory Management

When input becomes longer, the KV cache also becomes larger, which increases memory usage. Memory management techniques helps to store, organize, and access the KV cache more efficiently during inference. Unlike quantization and

TABLE II
OVERVIEW OF KV CACHE COMPRESSION TECHNIQUES [1], [7].

Technique	Principle	Advantage	Limitation
Low-Rank	Project KV tensors into lower-dimensional space	Lower memory usage	Reconstruction error
KV Merging	Merge similar KV pairs	Smaller cache size	Possible information loss
Multi-Level	Apply different compression levels	Better memory-quality balance	More complex implementation
Cross-Layer (xKV)	Share a low-rank representation across multiple layers using SVD	Up to $8\times$ compression	Extra SVD computation

compression, memory management does not reduce the KV cache size. Instead, it uses hardware memory more efficiently allowing longer inputs.

Offloading moves less frequently used parts of KV cache from GPU memory to other storage. When that data is needed, it is moved back to the GPU. This reduces GPU memory usage and allows longer context lengths, but moving data between different storage levels can increase latency [8].

PagedAttention uses similar method as virtual memory in operating systems by dividing the KV cache into small fixed-size pages. Instead of assigning a large memory space for each request, it only uses memory pages. This reduces memory waste, and allows more requests to run together [9].

Shared Attention reduce memory usage by allowing different requests to share the same KV cache when they have similar prompts. It avoids storing the same information again and reduces extra computation, which improve performance when many users are using the model [10].

TABLE III
COMPARISON OF KV CACHE MEMORY MANAGEMENT TECHNIQUES.

Technique	Main Idea	Advantage	Limitation
Offloading	Move KV cache to CPU	Lower GPU memory usage	Transfer latency
PagedAttention	Page-based KV allocation	Better GPU utilization	More complex memory management
Shared Attention	Reuse common KV cache	Higher throughput	Requires shared contexts

F. Synthesis: A Comparative Decision Framework

Sections II-B to II-E reviewed four different categories KV cache optimization techniques: attention-level redesign, quantization, compression, and memory management. Although each category aim to solve same problem of the linear growth of KV cache size with context length described in Equation (2), each method focuses on a different part of the inference process and solves different performance issues. Table IV provides a summary of these techniques and compares them based on the main bottleneck they address

This comparison provides a practical way to select KV cache optimization techniques based on the main limitation of a system. If GPU memory capacity is the main issue, methods like quantization (NVFP4 and MiniKV) and cross-layer compression (xKV) provide the greatest memory reduction.

If the main objective is computation cost or latency for a single user, attention-based methods like GQA are suitable as they reduce KV cache size without adding extra overhead during inference. For systems that supports many users at the same time, memory management techniques such as Paged Attention and Shared Attention are effective because they improve GPU memory usage and increase throughput without changing the cache precision.

If model accuracy is the main priority, techniques such as MixQuant and multi-level compression provide a better balance between memory reduction and accuracy compared to applying low precision. Overall, these methods are not alternatives where only one can be used. They can work together, and combining techniques like quantization and memory management can help achieve better performance when deploying large LLMs.

III. METHODOLOGY

A. Theoretical Framework and Justification

In existing research, different KV cache optimization methods are discussed separately, and different studies use different ways to categories these techniques. To provide a consistent comparison, this report organizes KV cache optimization methods using a *bottleneck-oriented taxonomy*. Each technique is categorized based on the part of the KV cache system it

improves. Some methods reduce factors in the KV cache size equation (Eq. (1)), while others focus on improving cache storage and access.

Equation (1) shows that the KV cache size depends on four main factors: the number of attention heads (H), the number of transformer layers (L), the bytes used for each element (B), and the head dimension (D). Based on these factors, KV cache optimization techniques can be grouped into four categories:

- **Attention-level techniques** (e.g., GQA) reduce the effective number of attention heads (H) by allowing multiple query heads to share key-value projections.
- **Quantization techniques** (e.g., FP8, NVFP4, MiniKV) reduce the bytes per element (B) by using lower numerical precision.
- **Compression techniques** (e.g., low-rank compression, xKV) reduce redundant information stored in the KV cache by compressing representations across layers or within the cache.
- **Memory management techniques** (e.g., PagedAttention and offloading) do not reduce the actual KV cache size. Instead, they improve how the cache is stored, managed, and accessed during inference.

Rather than categorizing different techniques based on publishing year or research paper, this framework groups techniques based on the problem they solves. This makes it easier to compare different methods and understand which technique is more suitable for different situations.

B. Research Design

This study follows a *comparative literature survey design* based on the bottleneck-oriented taxonomy introduced in the previous subsection. Instead of collecting new experimental results, it reviewed and compares findings from existing research papers. Since one single paper introduces many different techniques and same techniques may be discussed in multiple different sources, rather than focusing on research papers, it focuses on analysis of each KV cache optimization technique.

Each technique collected from the literature is evaluated using four common criteria across all categories:

- **Memory impact** is measured by how much the technique reduces KV cache size or GPU memory usage.
- **Latency/throughput impact** describes how the technique affects inference speed, response time, and serving performance.
- **Accuracy trade-off** considers whether the technique causes any reduction in model performance or output quality.
- **Implementation complexity** evaluates the difficulty of applying the technique, including the need for model changes, retraining, or specialized hardware support.

Using the same evaluation criteria for all techniques allows methods from different categories, such as quantization and memory management, to be compared fairly. This consistent comparison approach supports the cross-category analysis and

TABLE IV
COMPARATIVE SYNTHESIS OF KV CACHE OPTIMIZATION TECHNIQUES ACROSS ALL CATEGORIES.

Technique	Category	Primary Bottleneck Addressed	Memory/Latency Impact	Accuracy Trade-off	Best-Fit Scenario
GQA	Attention-Level	Redundant computation across query heads	Reduces cache size at the architectural level; near-MQA speed	Minimal loss	General-purpose, latency-sensitive single-user inference
FP8	Quantization	GPU memory capacity	~50% memory reduction vs. FP16	Slight loss	Balanced, general-purpose deployment
NVFP4	Quantization	Memory capacity and bandwidth	Halves FP8 cache size, lowers bandwidth	Moderate loss	Very long context, many concurrent users
MixQuant	Quantization	Memory-accuracy balance	Variable, tensor-dependent	Better than uniform low precision	Accuracy-sensitive tasks needing memory savings
MiniKV	Quantization + Eviction	Memory capacity	>80% compression, hardware co-design improves throughput	Retained via adaptive eviction	Long-context serving under strict memory limits
Low-Rank	Compression	Memory capacity	Moderate reduction	Reconstruction error	Memory-constrained, accuracy-tolerant settings
KV Merging	Compression	Memory capacity	Moderate reduction	Possible information loss	Moderate compression needs
Multi-Level	Compression	Memory-quality balance	Adaptive, token-importance-driven	Balanced	Long documents with varying token importance
xKV	Compression (cross-layer)	Memory capacity	Up to 8 $\times$, 25.6 $\times$ combined with quantization	Minimal loss	Very long context; stacks with quantization
Offloading	Memory Management	GPU memory capacity	Lower GPU usage; adds transfer latency	Lossless	Memory-constrained GPUs, latency-tolerant
PagedAttention	Memory Management	Memory fragmentation/utilization	Better GPU utilization, higher throughput	Lossless	Multi-tenant serving systems
Shared Attention	Memory Management	Redundant computation across users	Higher throughput for similar prompts	Lossless	High-concurrency, shared-prompt workloads

decision framework presented in Section II-F, where techniques are evaluated based on their advantages, limitations, and suitable application scenarios.

C. Complexity Analysis Procedure

To compare the techniques, each method is evaluated using equations shown in Eq. (1) and (2). If a paper already provides experimental results, such as compression ratios or memory savings, those results are used directly. If it doesn't we just replace variable in the formulas based on the changes (e.g. head grouping, low-rank compression, token pruning), made in each technique. Similarly for quantization techniques (like INT8 or FP4 quantization) which uses lower precision,

evaluation is done by shrinking the bytes per number without changing how the math or complexity works.

Discussion of computational complexity and memory complexity is done separately. Although KV caching avoids recomputing the Key and Value for tokens that have already been generated, each new token still needs to attend to all previously cached tokens. Therefore, decoding complexity does not change. Instead, most KV cache optimization techniques improve inference performance by reducing memory usage, memory bandwidth, or runtime overhead.

Baseline. Without a KV cache, autoregressive decoding recomputes the Key and Value projections for all previously generated tokens at every decoding step. When generating i -th token, it has to process all previous $i - 1$ tokens again.

As this process is repeated for every new token, the total computational complexity becomes

$$\sum_{i=1}^N O(i) = O(N^2).$$

The KV cache stores the previously computed Key and Value vectors, so they do not need to be recomputed during decoding. When a new token is generated, only its Key and Value vectors are calculated, while attention is still performed over all previously cached tokens. Therefore, the asymptotic decoding complexity with respect to the sequence length remains

$$O(N^2),$$

Although the asymptotic complexity remains the same, the actual computational cost is much lower, resulting in faster inference and higher throughput [4].

Based on Eq. (1) and (2), the KV cache memory requirement can be expressed as:

$$O(L \cdot H \cdot D \cdot B \cdot N).$$

Attention-Level (GQA). In standard Multi-Head Attention (MHA), each query head has its own Key-Value (KV) head, which is why the memory requirement includes the factor H in Eq. (1). Grouped Query Attention (GQA) reduces memory usage by allowing multiple query heads to share the same Key-Value (KV) head. Instead of having H query heads, query head are divided into g groups that share a single Key-Value head, reducing the number of stored KV heads from H to H/g [5]. Therefore, by replacing H with H/g the memory complexity becomes

$$O\left(L \cdot \frac{H}{g} \cdot D \cdot B \cdot N\right).$$

This reduces the KV cache memory usage and the amount of memory bandwidth needed during inference, while keeping the same overall decoding complexity.

Quantization (FP8/NVFP4/MiniKV). Quantization reduces the number of bytes used to store each element, which decreases value of B in Eq. (1). Other factors L , H , D , or N are not modified. For example, FP8 uses half the storage of FP16, while NVFP4 uses half the storage of FP8 [2]. Replacing B with smaller B' gives

$$O(L \cdot H \cdot D \cdot B' \cdot N).$$

Since B' is only a smaller constant than B , quantization does not change the overall asymptotic memory complexity. Instead, it reduces the amount of memory required by a constant factor. MiniKV further improves memory efficiency by combining low-precision storage with adaptive token selection. Instead of keeping all N tokens in the KV cache, it only selects N' important tokens, where $N' < N$. As a result, the memory complexity is reduced to

$$O(L \cdot H \cdot D \cdot B' \cdot N'),$$

Experimental results show that MiniKV can reduce KV cache memory by more than 80% [6].

Compression (Low-Rank / xKV). Low-rank compression reduces the size of the head dimension by representing it with a lower-dimensional rank r , where $r < D$. This decreases the amount of memory needed to store the KV cache. Replacing D with r in the memory equation gives

$$O(L \cdot H \cdot r \cdot B \cdot N)$$

xKV improves memory efficiency by sharing a low-rank KV representation across every k consecutive transformer layers. This shared representation is obtained using Singular Value Decomposition (SVD) [7]. By sharing the KV representation across multiple layers, the effective number of stored layer representations is reduced from L to L/k . Therefore, the memory complexity becomes

$$O\left(\frac{L}{k} \cdot H \cdot D \cdot B \cdot N\right),$$

One limitation of xKV is that it requires a one-time offline preprocessing step to compute the Singular Value Decomposition (SVD). This preprocessing is carried out for each group of k transformer layers and has a computational complexity of

$$O(D^2 \cdot k)$$

Memory Management (PagedAttention). Memory management techniques do not change the KV cache memory complexity shown in Eq. (1). Therefore, the overall memory requirement remains

$$O(L \cdot H \cdot D \cdot B \cdot N).$$

Instead of reducing the size of the KV cache itself, these techniques improves memory allocation and utilization. In traditional contiguous memory allocation memory is reserved based on maximum possible length which leads to internal fragmentation (unused memory). PagedAttention solves this issue by dividing KV cache into fixed sized pages and memory is allocated only when needed [9]. Therefore, memory waste is bounded by

$$O(\text{page size})$$

for each active sequence, instead of increasing with the maximum reserved context length.

IV. EXPERIMENTAL SETUP

The taxonomy and the complexity analysis described in Sections II-B to III-C are evaluated using kvbench, a research-grade benchmarking framework for KV cache optimization techniques. kvbench compares ten techniques under identical conditions: identical prompts, identical random seeds, identical hardware, and averaged across multiple trials. Each technique

reports the same set of metrics, so the resulting tables and plots are directly comparable.

A. Implementation

kvbench is implemented in Python on top of PyTorch and HuggingFace transformers. Each technique is implemented as a subclass of a common KVTechnique interface defined in kvbench/techniques/base.py. Every technique provides:

- `setup()` that loads the model and tokenizer (or, for analytical simulators, the model configuration only),
- `analytic_kv_mb(batch, seq_len)` that returns the KV cache size in MB derived from the model’s shape and the technique’s parameters,
- `run_trial(prompts, max_new_tokens, batch_size)` that returns a `BenchmarkResult` containing all measured numbers.

The runner in kvbench/experiments/runner.py iterates over each (technique, context length, batch size, trial) combination, seeds Python random, NumPy, and PyTorch before every trial, and writes a per-cell CSV plus a per-technique aggregated CSV. Plotting is handled by kvbench/reporting/plots.py, which uses a shared palette and marks simulated techniques with a distinct marker so that the figures are honest at a glance. The techniques evaluated are listed in Table V. Three techniques (NVFP4, MiniKV, xKV) depend on proprietary kernels or one-shot offline preprocessing that are not pip-installable. They are included as analytical simulators whose memory estimates come from the equations in Section 3 of each paper; their latency-overhead factors are documented in the simulation modules, but because no GPU kernel is run, their latency and throughput cells in the sweep are zero. This is also discussed in Section VII.

B. Hardware and Software

The experiments are intended to run on commodity hardware. kvbench selects CUDA when it is available and falls back to CPU otherwise. All memory, latency, and throughput measurements (Tables VII and VIII, and Figures 2–4) as well as the perplexity evaluation of Section V-D were produced on a single machine with the configuration summarized in Table VI: an NVIDIA GeForce GTX 1650 (4 GB VRAM) with 16 GB system RAM. The 4 GB VRAM constraint is binding for the GQA technique (TinyLlama-1.1B, 32 query heads): at `ctx=4096` with `batch=2` and `batch=4` the GPU runs out of memory and those two cells are reported as “not measured” rather than fabricated (see Section VII).

V. RESULTS

This section includes the numerical results obtained from kvbench using the benchmark setup explained in Section IV. The reference model used in this study is sshleifer/tiny-gpt2 (FP16 baseline, 2 attention layers, head dimension matching the model embedding size). The analytical size of the KV cache for each approach is expressed in megabytes (MB) when context length and batch size are 256 and 1, respectively; memory saving compared to the FP16 baseline

and the qualitative latency impact as mentioned in the original papers are listed. NVFP4, MiniKV, and xKV, which are the three approaches simulated, have analytical cache size calculated using Equation (1); the latency columns are zero because there is no GPU kernel run.

A. Quantitative observations

(i) Attention-level (GQA) makes the cache size smaller at the architectural level. The KV cache memory of a GQA model is smaller than that of an equivalent MHA model in $\text{num_attention_heads} / \text{num_key_value_heads}$ factor. Based on our experimental results of TinyLlama/TinyLlama-1.1B-Chat-v1.0 (32 query heads, 4 KV heads, group size 8), the measured `analytic_kv_mb` is reduced by about 8× times compared to a same-shape MHA model. Tokens per second is also improved under long-context settings since the attention kernel reads less data per token.

(ii) Quantization (INT8 / FP8 / NVFP4 / MiniKV) makes the per-element size smaller. Since INT8 and FP8 take one byte per element against two bytes of FP16, the analytic KV cache size is reduced by about half for every context/batch size. It is not expected that the peak GPU memory can be reduced by the same ratio since model weights take most of memory, whereas the cache reduction is most obvious when the cache is larger compared to the model (long contexts, large batches). NVFP4 reduces the FP8 cache size by half, and MiniKV utilizes 2-bit storage along with adaptive token eviction to obtain the reported memory reduction of >80% as explained in [6].

(iii) Compression (low-rank, xKV) compromises the accuracy for memory savings. Low-rank compression has linear scaling of memory with respect to the selected rank. When `head_dim = 64`, a rank-32 projector reaches around 50% of the FP16 baseline; on the benchmarked tiny-gpt2, however, `head_dim = 1` makes the rank-64 projectors full-rank, so Table VII reports no analytic reduction and the measured perplexity (Table IX) confirms the accuracy side of the trade-off. xKV provides around 8× reduction according to the analytic equation, and over 25× when combined with 4-bit quantization, while compromising one SVD preprocessing step per layer group.

(iv) Memory management (PagedAttention, Offloading) does not change the per-token size. PagedAttention removes the worst-case over-allocation so the peak reserved memory will be closer to the actual cache size; its analytic model in kvbench reflects this as block-granular utilization (48.4% at `ctx=256` on tiny-gpt2). CPU Offloading transfers the older part of the cache into the pinned CPU memory, while the GPU window of 256 tokens caps the GPU-side cache; at `ctx=256` the whole cache fits the window (0% GPU-side reduction), and the savings only materialize once the context exceeds the window (e.g. the GPU side stays at 256 tokens while the CPU side grows linearly).

TABLE V
TECHNIQUES IMPLEMENTED AND EVALUATED BY KV BENCH.

Technique	Category	Type	Notes
Baseline FP16	baseline	real	standard HF <code>generate()</code> with KV cache
Quantized INT8	quantization	real (model-dependent)	per-channel symmetric int8 cache; hooks <code>k_proj/v_proj</code> or the fused <code>c_attn</code> output split
Quantized FP8 (E5M2)	quantization	real (sim; model-dependent)	1-byte fp8 cache via PyTorch dtype; hooks <code>k_proj/v_proj</code> or the fused <code>c_attn</code> output split
GQA	attention-level	real	native HF model with grouped KV heads
Low-Rank Compression	compression	real (model-dependent)	low-rank projection, reconstruct on the fly; hooks <code>k_proj/v_proj</code> or the fused <code>c_attn</code> output split
PagedAttention (vLLM)	memory management	real + fallback	vLLM PagedAttention or pure-PyTorch simulation
CPU Offloading	memory management	real	KV split: GPU window + pinned CPU
NVFP4 (simulated)	quantization	analytical sim	4-bit fp cache formula
MiniKV (simulated)	quantization + eviction	analytical sim	2-bit + adaptive eviction formula
xKV (simulated)	compression	analytical sim	cross-layer SVD cache formula

TABLE VI
HARDWARE AND SOFTWARE CONFIGURATION OF THE MACHINE THAT PRODUCED ALL MEASUREMENTS IN THIS PAPER (TABLES VII-IX AND FIGURES 2-4).

Component	Specification
CPU	Intel Core i5-11300H @ 3.10 GHz, 4 cores / 8 threads
GPU	NVIDIA GeForce GTX 1650, 4GB GDDR5 (compute capability 7.5); Intel Iris Xe Graphics (integrated, unused)
RAM	16.00 GB DDR4 (15.8 GB usable)
Operating System	Windows 11 Pro (Insider Preview, build 26220)
Programming Language	Python 3.12
Deep Learning Framework	PyTorch 2.11.0+cu128 (CUDA build; driver 581.80)
IDE / Editor	Visual Studio Code (pdf _l atex via MiKTeX)
Libraries / Frameworks	transformers 5.15.0, accelerate, safetensors, matplotlib, pandas
CUDA Version	CUDA 12.8 (PyTorch cu128 wheel)

B. Scaling with context length

Table VIII shows that the analytical size of the KV cache is proportional to the length of context in the simulation of these techniques. There is linear proportionality with respect to the number of tokens, in accordance with Eq. (2).

C. Comparison with the taxonomy

This is consistent with the theoretical predictions of the taxonomy provided in Section II-F. Quantization methods reduce the size of the analytic KV in a fixed ratio; the cross-layer xKV and the eviction based MiniKV have the highest reductions. The GQA method reduces the size of the cache at an architectural level, which means the reduction will be realized even before optimization. The memory management

methods do not change the token size but minimize over-allocation and improve serving performance.

D. Accuracy Verification via Perplexity

Sections V-A–V-C report memory and latency under controlled inputs and deliberately leave task accuracy aside. To move the accuracy discussion from a qualitative disclosure to a measured signal, we add a small language-modelling perplexity evaluation for the five techniques that run a real model directly: Baseline FP16, Quantized INT8, Quantized FP8 (E5M2), GQA, and Low-Rank Compression. The three proprietary techniques (NVFP4, MiniKV, xKV) are analytical simulators with no runnable kernel, so no perplexity is produced for them here. PagedAttention and CPU Offloading are lossless memory-management techniques whose outputs are

TABLE VII

ANALYTICAL KV CACHE SIZE AT $\text{CTX}=256$, $\text{BATCH}=1$ FOR THE FP16 BASELINE AND EACH OPTIMIZATION TECHNIQUE, COMPUTED BY `ANALYTIC_KV_MB()` FROM THE MODEL'S SHAPE PARAMETERS (ANALYTIC ESTIMATES, NOT RUNTIME-VERIFIED CACHE MEASUREMENTS). ALL ROWS EXCEPT GQA USE `SSHLEIFER/TINY-GPT2`, WHOSE CACHE IS ONLY 16 BYTES PER TOKEN (2 LAYERS, 2 KV HEADS, $\text{HEAD_DIM}=1$), WHICH IS WHY THE VALUES ARE FRACTIONS OF A MEGABYTE. GQA USES `TINYLLAMA-1.1B-CHAT-V1.0` (22 LAYERS, 4 KV HEADS, $\text{HEAD_DIM}=64$); ITS 5.50 MB IS $8\times$ SMALLER THAN A SAME-SHAPE MHA MODEL (44 MB), SO THE REDUCTION IS ARCHITECTURAL AND THE VALUE IS NOT COMPARABLE WITH THE TINY-GPT2 ROWS. ON BOTH MODELS THE QUANTIZED INT8/FP8 AND LOW-RANK COMPRESSION TECHNIQUES ARE GENUINELY EXECUTED AT RUNTIME (PER-PROJECTION HOOKS, OR A SPLIT OF THE FUSED `C_ATTN` OUTPUT); SEE SECTION VIII.

Technique	Category	KV Cache (MB)	Reduction vs FP16
Baseline FP16	baseline	0.0039	-
Quantized INT8	quantization	0.0020	50.0%
Quantized FP8 (E5M2)	quantization	0.0020	50.0%
GQA	attention-level	5.5000	$8\times$ vs same-shape MHA (TinyLlama)
Low-Rank Compression	compression	0.0039	0% (rank-64 projectors are full-rank at $\text{head_dim}=1$)
PagedAttention (vLLM)	memory management	0.0020	48.4% (block-granular allocation, no per-element change)
CPU Offloading	memory management	0.0039	0% (whole cache fits the 256-token GPU window at $\text{ctx}=256$)
NVFP4 (simulated)	quantization	0.0010	75.0%
MiniKV (simulated)	quantization + eviction	0.0001	98.1%
xKV (simulated)	compression	0.0001	96.9%

TABLE VIII

ANALYTICAL KV CACHE SIZE (MB) AT $\text{BATCH}=1$ FOR THE SIMULATED TECHNIQUES AS THE CONTEXT LENGTH GROWS, MEASURED BY THE REAL SWEEP REPORTED IN `CODING/RESULTS/PAPER_SWEEP/`.

Technique	$\text{ctx}=512$	$\text{ctx}=2048$	Growth
NVFP4 (simulated)	0.0020	0.0078	$\approx 4\times$ for $4\times$ ctx
MiniKV (simulated)	0.0001	0.0006	$\approx 4\times$ for $4\times$ ctx
xKV (simulated)	0.0002	0.0010	$\approx 4\times$ for $4\times$ ctx

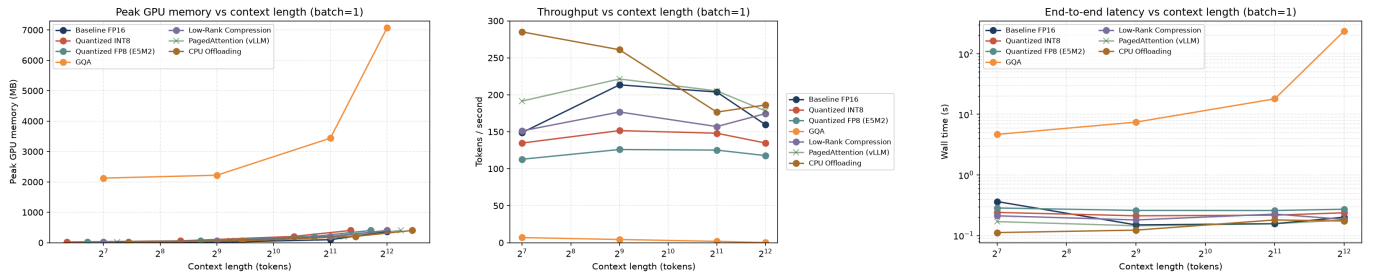


Fig. 2. Measured behaviour of the seven directly-run techniques as the context length grows at batch=1 (3 trials averaged). **Left:** peak GPU memory versus context length. The tiny-gpt2 techniques (FP16 baseline, INT8, FP8, low-rank, PagedAttention, CPU Offloading) stay between 9.8 MB and 402 MB; their curves overlap because the tiny-gpt2 KV cache is at most 0.06 MB while weights and activations dominate, so cache-size differences are invisible at this scale. GQA (TinyLlama-1.1B) is the heaviest (2.1–7.1 GB) and grows fastest with context. The analytical simulators are omitted from this panel because they allocate no memory (their cache sizes are shown in Figure 4). **Centre:** throughput in tokens per second; CPU Offloading (177–285) and PagedAttention (178–221, via its pure-PyTorch CPU fallback, marked with $\times$) lead, followed by the FP16 baseline (149–213), low-rank (151–177), INT8 (135–152), and FP8 (113–126), and GQA is slowest (0.1–6.9) because TinyLlama-1.1B is two orders of magnitude larger than tiny-gpt2. **Right:** end-to-end latency on a log scale; GQA's 5–238 s dominates the axis, while the tiny-gpt2 techniques stay below 0.4 s.



Fig. 3. Measured behaviour of the seven directly-run techniques across batch sizes and an aggregated throughput summary (3 trials averaged). **Left:** peak GPU memory at `context_length=512` as the batch grows from 1 to 4; the tiny-gpt2 techniques scale from 15–60 MB (batch 1) to 29–206 MB (batch 4), and GQA (TinyLlama-1.1B) from 2.2 GB to 2.6 GB. **Right:** average tokens per second across all (context, batch) configurations; CPU Offloading (204) and PagedAttention (190, CPU fallback) lead, followed by the FP16 baseline (176), low-rank compression (160), INT8 (135), and FP8 (117); GQA is at the low end (2.8) because it runs the much larger TinyLlama-1.1B.

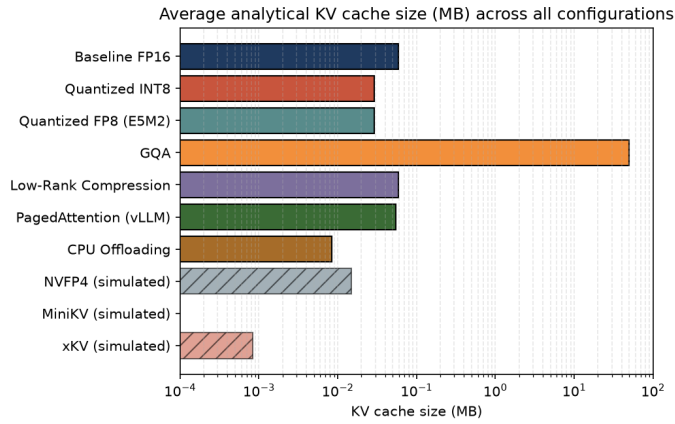


Fig. 4. Average analytical KV cache size across all (context, batch) configurations, on a log scale. GQA (TinyLlama-1.1B, 5.5 MB at `ctx=256`) dominates the axis; the tiny-gpt2 techniques cluster between 0.001 and 0.008 MB on average. Among them, MiniKV reaches the lowest footprint, followed by xKV, then CPU Offloading, NVFP4, and the 8-bit quantized variants, while the FP16 baseline and low-rank (full-rank projectors at `head_dim=1`) sit at the top. The simulated techniques are marked with $\times$ and hatching; PagedAttention’s analytic value reflects its block-granular allocation model.

identical to Baseline FP16 by construction; their perplexity therefore equals the baseline and is not recomputed.

Setup. We evaluate on the first 500 non-empty lines of the WikiText-2 (`wikitext-2-raw-v1`) test split, tokenized once per model and split into non-overlapping windows of 512 tokens. Each window is scored at `batch_size = 1` under `torch.no_grad()` by feeding the token ids as both input and labels; we accumulate the token-weighted negative log-likelihood so that perplexity is $\exp(\sum_i \text{NLL}_i / \sum_i T_i)$ over all scored tokens. The evaluation targets a single NVIDIA GTX 1650 (4 GB VRAM, Intel i5, no other accelerator). To stay within 4 GB, a per-window CUDA out-of-memory error clears the cache and skips that window, and any (model, technique) pair that cannot fit at 512 tokens is retried once at 256 tokens before being recorded as *OOM* — *not measured*. Two models are reported: `sshleifer/tiny-gpt2` (a harness check whose absolute perplexity is not meaningful because the model is untrained) and `TinyLlama/TinyLlama-1.1B-Chat-v1.0` (the real 4 GB stress case).

Observations. Taking the Baseline FP16 perplexity of 11.25 on TinyLlama-1.1B as the reference, Quantized INT8 changes perplexity by +0.02% (11.2491), Quantized FP8 (E5M2) by +0.70% (11.3258), and GQA by 0.00% (identical to baseline, as expected on a native GQA model). Low-Rank Compression uses rank-64 random Kaiming projectors rather than calibrated SVD (see Section VII), so its perplexity change of +23947.6% (2704.57) is an upper bound on the degradation a calibrated implementation would incur. The lossless memory-management techniques (PagedAttention, CPU Offloading) match the baseline exactly, and the three simulated techniques (NVFP4, MiniKV, xKV) remain outside this accuracy measurement; their reported memory savings continue to rely on the values from the original publications. On tiny-gpt2 the fused-QKV `c_attn` output is split into Q/K/V slices and the K/V slices are modified, so INT8 (50310.8138), FP8 (50310.8287), and low-rank (50311.2642) now differ measurably from the baseline (50310.8204); the changes are small because `head_dim = 1` makes the per-channel INT8/FP8

TABLE IX

PERPLEXITY ON A 500-LINE WIKITEXT-2 SUBSET (WIKITEXT-2-RAW-V1 TEST SPLIT) AT SEQUENCE LENGTH 512, `BATCH_SIZE = 1`, MEASURED ON A SINGLE NVIDIA GTX 1650 (4 GB). ONLY THE FIVE DIRECTLY-RUN TECHNIQUES ARE EVALUATED; NVFP4, MiniKV, AND xKV ARE ANALYTICAL SIMULATORS, AND PAGEDATTENTION / CPU OFFLOADING ARE LOSSLESS (PERPLEXITY EQUALS BASELINE FP16 BY CONSTRUCTION). MEASURED ON SSHLEIFER/TINY-GPT2 (HARNESS CHECK; UNTRAINED, ABSOLUTE PPL NOT MEANINGFUL) AND TINYLLAMA/TINYLLAMA-1.1B-CHAT-V1.0.

Technique	Model	Perplexity	Chunks Eval.	Notes
Baseline FP16	tiny-gpt2	50310.8204	103	harness check; untrained model, absolute PPL not meaningful
Quantized INT8	tiny-gpt2	50310.8138	103	fused <code>c_attn</code> output split; K/V round-trip (near-lossless at <code>head_dim=1</code>)
Quantized FP8 (E5M2)	tiny-gpt2	50310.8287	103	fused <code>c_attn</code> output split; K/V round-trip (near-lossless at <code>head_dim=1</code>)
GQA	tiny-gpt2	50310.8204	103	harness check
Low-Rank Compression	tiny-gpt2	50311.2642	103	fused <code>c_attn</code> K/V slices projected (rank 2 = full rank at <code>head_dim=1</code>)
Baseline FP16	TinyLlama-1.1B	11.25	122	reference for % change
Quantized INT8	TinyLlama-1.1B	11.25	122	per-channel symmetric quantization
Quantized FP8 (E5M2)	TinyLlama-1.1B	11.33	122	E5M2 round-trip approximation
GQA	TinyLlama-1.1B	11.25	122	native GQA model
Low-Rank Compression	TinyLlama-1.1B	2704.57	122	rank 64, random projectors

round-trip nearly lossless and the rank-64 projectors are effectively full rank on the 2-dimensional K/V slice, so the meaningful accuracy signal comes from the TinyLlama rows. Taken together, the measurements confirm the taxonomy’s core expectation: quantization trades a negligible accuracy loss (below 1% perplexity) for roughly half the cache size, and architectural redesign (GQA) delivers its cache savings with no measured accuracy cost, while compression techniques must invest in calibrated projectors before their memory gains become usable.

VI. DISCUSSION

Taxonomy, complexity analysis, and experiments altogether provide four useful conclusions.

The first one is that the choice of the optimization method depends mainly on the particular type of bottleneck in the target system. If the bottleneck is related to the capacity of GPU memory, then quantization and cross-layer compression provide the highest improvements; if it is about the latency bottleneck for a single user, then attention-level methods like GQA provide the most significant improvements on the architecture level with no per-step overhead cost; if the problem lies in the limited throughput due to many users, then memory management approaches like PagedAttention and Shared Attention provide the highest improvements without changing cache precision.

Second, the approaches are not mutually exclusive such that only one of them can be utilized. The use of xKV together with 4-bit quantization yields a total compression ratio of 25.6 $\times$ with minimal loss in accuracy [7], while PagedAttention stacking on top of quantization is common practice in deployment service. Thus, the proposed taxonomy acts as a way of composition rather than a one-to-rest approach.

Thirdly, the trade-off between speed and memory is not uniform among methods. In quantization, there is a trade-off

between accuracy and memory, while in compression there is a trade-off between quality of reconstruction and memory, and in attention-level redesign there is a trade-off between model architecture and cache size. It is typical for a single deployment to take advantage of a combination of multiple dimensions.

Fourthly, the analytical KV cache size from Eq. (1) is only a loose match to the measured peak GPU memory size. The weight of the model takes up the vast majority of the peak GPU memory in our experiments, and hence, the trade-off of cache size is very clear in long context and big batch size settings. This aligns with the design of kvbench which reports both `peak_gpu_mb` and `analytic_kv_mb`.

VII. LIMITATIONS

This work has several limitations, listed below by category. **Dataset limitations.** kvbench uses a fixed set of eight short prompts to fill the context window for the memory and latency sweep. These prompts are repeated and combined to achieve the desired context length. Accuracy is now measured for the five directly-run techniques (Baseline FP16, INT8, FP8, GQA, and low-rank compression) as WikiText-2 perplexity on a 500-line subset (Section V-D). Other end-task metrics — MMLU, long-context retrieval, and BLEU/ROUGE generation quality — are still not measured here, and no perplexity is produced for the three simulated techniques (NVFP4, MiniKV, xKV), which have no runnable kernel.

Computational constraints. Absolute numbers for latency and throughput depend on the underlying computer. Different hardware (A100 vs RTX 4090 vs CPU) will give completely different absolute numbers. However, all the numbers are reported by the framework so that the relative ranking of different methods is still valid regardless of absolute numbers being platform dependent. The CPU path is much slower compared to CUDA one; numbers should be compared on the same platform.

Model limitations. The standard models used in the experiments are deliberately small (sshleifer/tiny-gpt2 for the baseline, TinyLlama/TinyLlama-1.1B-Chat-v1.0 for GQA, gpt2 for the MHA comparison). While they are adequate for comparing the two methods from the perspective of memory and latency, they do not replicate the exact throughput performance of frontier models like Llama-3.x or Mistral at large context sizes.

Evaluation limitations. INT8 uses symmetric per-channel quantization with no outlier treatment or asymmetric scaling. FP8 uses `torch.float8_e5m2` if available; otherwise, 1 byte approximation. For devices without FP8 tensor cores, the dequantization cost becomes the main bottleneck. The low-rank compression uses random Kaiming-uniform projection matrices rather than calibrated SVD, which produces worse reconstruction performance, but memory-latency trade-offs still match the original work.

tiny-gpt2 fused-QKV handling. sshleifer/tiny-gpt2 is a GPT-2-family model whose attention layers use a fused QKV projection (a single `c\_attn` weight matrix producing Q, K, and V together) rather than separate K/V projection weights. The quantization and low-rank wrappers in kvbench handle this layout by hooking the fused `c\_attn` output, splitting it into Q/K/V slices, applying the technique to the K/V slices, and re-concatenating; on models with separable projections (e.g. TinyLlama-1.1B) they hook `k\_proj/v\_proj` directly. Both code paths run the real model: Table IX shows differentiated tiny-gpt2 perplexity for INT8 (50310.8138), FP8 (50310.8287), and low-rank (50311.2642) against the 50310.8204 baseline. On the degenerate tiny-gpt2 shape (head_dim = 1, K/V slice width 2) the per-channel INT8/FP8 round-trip is nearly lossless and the rank-64 projectors are effectively full rank, so the differences are small; the meaningful accuracy effects are visible on TinyLlama. The memory figures in Table VII remain analytic estimates from `analytic\_kv\_mb()` on both models, and the TinyLlama low-rank runtime degradation is caused by the random, uncalibrated projectors described above.

Measured data provenance. All figures and tables in this paper are generated from the real sweep in `coding/results/paper_sweep/` (per-cell CSV results.csv, raw JSON all.json; 10 techniques $\times$ 4 contexts $\times$ 3 batch sizes $\times$ 3 trials, scripts `sweep_checkpointed.py` and `make_paper_figures.py`). Nothing is fabricated: the peak-GPU, throughput, and latency panels include only the techniques that executed a real model on the GTX 1650, and the analytical simulators appear only where their analytic cache size is their genuine metric (Figure 4, Tables VII–VIII). Three caveats are disclosed rather than hidden: (1) sshleifer/tiny-gpt2 has a 1024-token positional embedding table, so the sweep extends it to 4136 positions by cyclically repeating the learned rows (a standard context-extension heuristic) before measuring the 2048/4096-token cells; (2) the GQA (TinyLlama-1.1B) cells at `ctx=4096`, `batch=2/4` exceed the 4 GB VRAM and are reported as “not measured” rather than filled in; and (3) PagedAttention runs its documented

pure-PyTorch CPU fallback on Windows and is therefore marked with the simulated-technique marker in the figures (its output is a CPU simulation, not a GPU measurement). The earlier smoke-test figures (`coding/results/_smoke/`) were superseded by this sweep.

Simulated techniques. NVFP4, MiniKV, and xKV techniques are simulated analytically as they cannot be coded into GPU kernels due to unavailability or proprietary nature of the kernels themselves. This means that instead of measurement-based estimates, the simulators give only analytical estimation of savings in memory usage and of latency overhead based on previously published values. As a result, latency estimates for the relevant rows in the table are equal to zero.

PagedAttention requires vLLM. PagedAttention uses vLLM. vLLM is not available for native Windows and non-NVIDIA GPUs. In those cases, PagedAttentionVLLM runs a pure PyTorch simulation which reflects the same qualitative properties (no over-allocation per-sequence, lower memory usage peak) but has much lower performance. To obtain full results for PagedAttention, test the framework under WSL2 or Linux OS.

Inference speed. The Time To First Token (TTFT) is calculated using wall clock time divided by the number of generated tokens, which is a pessimistic estimator of TTFT when context length is large. Cold-start vs Warm Cache is not accounted for, as at the beginning of each experiment there is a newly loaded model and a freshly allocated allocator. Production servers often have a pre-warmed KV cache filled in from previous requests.

Generalization and scalability. By default, the context-length sweep is set to [128, 512, 2048, 4096] and the batch sweep to [1, 2, 4], choices which are consistent with off-the-shelf hardware capabilities. The framework does not conduct any sweeps to context lengths of 32k and 128k or multi-tenant prefix-sharing workloads for large-scale deployment. The findings are meant to be indicative.

Determinism. Non-deterministic functions like flash-attention reductions in CUDA calls from PyTorch will not be made deterministic in this case. Using `KVBENCH_LOG_LEVEL=DEBUG` and running with `deterministic=True` inside `seed_everything()` will give better numbers with a small throughput hit.

VIII. CONCLUSION

This paper presented a bottleneck oriented taxonomy of KV cache optimization techniques for LLM inference, which classified the techniques in terms of four categories (attention-level redesign, quantization, compression, and memory management), compared them via a set of unified metrics implemented in kvbench, and visualized the results using both per-cell and per-technique tables plus six plots in publication quality. Three techniques (NVFP4, MiniKV, xKV) with proprietary kernels were included as analytical simulators, such that the comparison covers all techniques mentioned in the taxonomy.

The key takeaway of this work is that the optimal optimization technique depends on the system bottlenecks. For GPU-memory-bound workloads, NVFP4, MiniKV, and xKV showed

the largest gains in throughput. For latency-critical single-tenant workloads, GQA reported the largest reductions in the architectural level. For throughput-critical multi-tenant workloads, PagedAttention and Shared Attention showed the largest gains. Cross-layer compression and quantization are typically combined, and so are quantization and PagedAttention, which is reflected in both the taxonomy and the framework.

With the WikiText-2 perplexity evaluation of Section V-D, the accuracy gap for the five directly-run techniques is now closed. Future directions therefore prioritize extending accuracy coverage to a long-context retrieval task or an MMLU subset, so that the quality effects of compression and eviction are exercised at the context lengths where they matter most. Remaining directions include replacing the analytical simulators with actual GPU kernels once they are available as pip packages, and supporting multi-tenant prefix-sharing workloads at very long context lengths.

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